SYEDA MALIHA TABASSUM

syeda.maliha.tabassum1@g.bracu.ac.bd | linkedin.com/in/syeda-maliha-tabassum19

Computer science graduate and researcher working at the intersection of multimodal AI, agentic systems, and reliable decision-making. My work focuses on human-centered sensing, assistive intelligence, edge–cloud AI systems, and automation, with growing interest in AI evaluation, calibration, and sequential decision-making under uncertainty. I am particularly interested in building and evaluating AI systems that can support robust real-world decisions in high-stakes and human-centered environments.

EDUCATION

B.Sc in Computer Science and Engineering, BRAC University — Sep 2022 – Sep 2026
CGPA: 3.93 / 4.00

Higher Secondary Certificate, Birshreshtha Noor Mohammad Public College — Jul 2019 – Feb 2022
GPA: 5.00 / 5.00

RESEARCH INTERESTS

Reliable AI Evaluation, AI Calibration, Sequential Decision-Making, Agentic AI Systems, Multimodal Learning, Human-Centered AI, Edge–Cloud AI Systems, Automation, Assistive Intelligence

PUBLICATION

Vision-based Hand Gesture Virtual Keyboard-Mouse Framework with Bilingual Next-word Prediction 2026
Sammam Mahdi, Reshad Ul Karim, Tareq Sujat, **Syeda Maliha Tabassum**, Abrar Samin, and Aniqua Nusrat Zereen
Proc. 2026 23rd International Joint Conference on Computer Science and Software Engineering (JCSSE), IEEE, 2026, pp. 638–643. doi:10.1109/JCSSE68839.2026.11596610

RESEARCH EXPERIENCE

Less Than Meets the Eye: Action-Set Identifiability and Principled Abstention for 3D Physical Geometry
Under Review: ICLR 2027 Investigated whether available camera motions can distinguish competing physical explanations of a scene, with principled abstention when the scene's 3D physical geometry remains unidentifiable.

CAMDet: Camera-Agnostic Monocular Metric 3D Object Detection
Under Review: IEEE Transactions on Image Processing Developed a monocular 3D object-detection framework with learned intrinsic feature conditioning, designed to generalize across cameras without retraining. Integrated metric-depth and uncertainty estimation to support real-time edge inference.

Beyond Discrimination: Toward Trustworthy Autism Classification from Resting-State Functional Connectomes through Calibration and Interpretable Modeling
Under Review: Computers in Biology and Medicine Evaluated autism classifiers using resting-state functional connectivity, examining discrimination, calibration, cohort effects, and explanation stability to assess reliability and interpretability.

Vision-Based Fall Detection: Systematic Review of Methods, Trends, and Challenges
Under Review: Artificial Intelligence in Medicine Systematically reviewed 121 vision-based fall detection studies across skeleton, appearance, and hybrid approaches, assessing real-world validity, robustness, deployment feasibility, and evidence quality.

LUMENAA – Language-Unified Multimodal Edge-Native Agent for Assistive Perception In Progress
Developing an agentic multimodal system integrating vision, language, and temporal memory within a hybrid edge–cloud architecture to support hierarchical planning, adaptive perception, and low-latency decision-making.

Multimodal Risk Prediction in Urban Road Scenes In Progress
Developing a multimodal spatial–temporal framework integrating segmentation, pose estimation, and engineered spatial features for pedestrian risk prediction, emphasizing interpretable modeling and uncertainty-aware decision pipelines.

An Upstream-Routed Adaptive Pipeline for Non-Invasive PCOS Screening and Phase Tracking In Progress
Developing a two-stage, condition-aware architecture that uses an explainable non-invasive PCOS classifier to adapt downstream time-series tracking filters, aiming to reduce tracking failures and false ovulation spikes in high-variance cycles.

Robust Follicle Segmentation in 2D Ultrasound via Low-Rank Adapted MedSAM In Progress
Developing a parameter-efficient MedSAM framework using SRAD/CLAHE preprocessing and LoRA for few-shot ovarian follicle segmentation in noisy 2D ultrasound images to support PCOS assessment.

WORK EXPERIENCE

NEXTERN Jun 2026 – Sep 2026
Technology Infrastructure Management, Technology, Grameenphone Ltd.
Automated rectifier and Li-Ion battery deployment and connectivity monitoring and weekly reporting using Huawei NetEco. Coordinated SMU upgrades with the Umbrella team and prepared requirements for an AI-assisted site power design tool.

Undergraduate Research Assistant – DRISTEE Oct 2025 – Oct 2026
BRAC University — RSGI Grant 2025
Conducted analysis and experiments for an assistive technology project, supporting solution development and evaluation. Prepared technical documentation and presentations and coordinated research activities within a collaborative team.

Undergraduate Teaching Assistant – Data Structures Feb 2025 – Jan 2026
Department of Computer Science and Engineering, BRAC University
Mentored 250+ students through laboratory sessions and individual guidance, explaining programming concepts and supporting problem solving. Evaluated assignments and supported course activities to maintain consistent academic delivery.

Core Team Member – Vision, Networking, and Mechanical Systems Sep 2023 – Jul 2025
BRAC University Mars Rover Project (Mongoltori)
Contributed to development, testing, and technical troubleshooting across vision, networking, and mechanical subsystems. Collaborated with multidisciplinary teams to support system integration and project execution.

PROJECTS

Arduino-Based Multi-Sensor Health Monitoring and Fall Detection June 2025 - Sep 2025
Developed an embedded framework integrating biomedical sensors with sensor fusion and microcontroller coordination. Implemented communication and alert pipelines using GSM/GPS and I^2C for wearable health monitoring. (Report)

Autonomous ArUco Marker–Based Multi-Terrain Equipment Service Bot Oct 2025 - Jan 2026
Built a distributed mobile manipulation system integrating ArUco-based 6-DoF localization, autonomous navigation, and PBVS control within a dual-ESP32 architecture. (GitHub Repository)(video)

WeHeal Feb 2025 - Sep 2025
Designed a distributed MERN-based healthcare system with modular architecture. Developed interoperable pipelines for telemedicine, scheduling, secure transactions, and digital health services, emphasizing scalability. (GitHub Repository) (Website)

PathGuard: Discrete Logic-Driven Obstacle Avoider Feb 2025 - May 2025
Built a microcontroller-free autonomous navigation system using discrete logic circuits and sensor-driven control to study low-level robotic decision-making. (GitHub Repository)

Predictive Modeling of Smoking Behavior from Biomedical Data Feb 2025 - May 2025
Developed a biomedical machine learning pipeline with feature engineering and multi-model evaluation to predict smoking behavior, emphasizing robust validation and interpretability. (GitHub Repository)

TECHNICAL SKILLS

Programming & Algorithms: Python, C/C++, Java, SQL, JavaScript, Data Structures, Algorithmic Problem Solving
Machine Learning & AI: PyTorch, Scikit-learn, Deep Learning, Vision Transformers, Multimodal Learning, Feature Engineering, Model Evaluation, Uncertainty-Aware Modeling, Statistical Modeling, Optimization Techniques, Data-Driven Decision
Computer Vision & Human-Centered AI: YOLO, OpenCV, MediaPipe, Pose Estimation, Vision–Language Pipelines, Assistive AI Systems, Human Activity and Gesture Recognition
Agentic Systems & Decision Support: Agentic AI Systems, Sequential Decision Pipelines, Human-Centered Sensing, Edge–Cloud AI Architectures, Low-Latency Inference
Automation, Robotics & Embedded Systems: Infrastructure Automation, Embedded Systems, ESP32, Arduino, Sensor Fusion, Communication Networks, GSM/GPS, I^2C, HAM Radio
Tools & Development: Git, Linux, LaTeX, OpenGL, MERN Stack, REST APIs, Technical Documentation

HONORS AND AWARDS

Dean's List and Vice Chancellor's List (BRAC University) Jan 2023 – Sep 2026
Merit-Based Academic Scholarship (BRAC University) Jan 2023 – Sep 2026
Top 6 Finalist — EXCELERATE National Excel Competition (FINACT, BRAC University) Dec 2025
1st Runner-Up — CSE471 Project Demonstration (BRAC University Computer Club) Dec 2025
Second Runner-Up — RoboSoccer Competition-TechSpectra (Robotics Club of BRAC University) May 2024
Semi-Finalist — RoboSoccer Competition-TRACTION (Robotics Club of BRAC University) Apr 2025
Semi-Finalist — Hult Prize On-Campus Round (BRAC University) Feb 2023
Bronze Level Award — The Duke of Edinburgh's International Award May 2023 – Mar 2024